

UNIVERSITY OF TECHNOLOGY

DATA STRUCTURES & ALGORITHMS - COURSE STRUCTURE

Semester: 7

Instructor: Sunil Sharma (CSE)

Total Marks: 100

Marks Distribution

Component	Marks
Mid Semester Exam	20
End Semester Exam	40
Assignments (4)	10
Quizzes (4)	10
Lab / Coding Tests	10
Attendance	5
Viva / Project	5
Total	100

Course Plan

Unit 1 (Week 1-2)

- 1 Introduction to DSA
- 2 Time & Space Complexity
- 3 Big-O, Omega, Theta
- 4 Recursion Basics
- 5 Quiz 1 (5 Marks)
- 6 Assignment 1 (2.5 Marks)

Unit 2 (Week 3-4)

- 1 Arrays
- 2 Linked Lists
- 3 Coding Test 1 (5 Marks)
- 4 Assignment 2 (2.5 Marks)

Unit 3 (Week 5-6)

- 1 Stacks
- 2 Queues
- 3 Expression Evaluation
- 4 Quiz 2 (5 Marks)

Mid Semester Exam (Week 7)

- 1 Covers Unit 1-3
- 2 Marks: 20

Unit 4 (Week 8-9)

- 1 Binary Trees
- 2 BST
- 3 Heaps
- 4 Assignment 3 (2.5 Marks)
- 5 Coding Test 2 (5 Marks)

Unit 5 (Week 10-11)

- 1 Graphs

- 2 BFS, DFS
- 3 Dijkstra, MST
- 4 Quiz 3 (5 Marks)

Unit 6 (Week 12-13)

- 1 Dynamic Programming
- 2 Greedy
- 3 Backtracking
- 4 Assignment 4 (2.5 Marks)
- 5 Quiz 4 (5 Marks)

End Semester Exam (Week 14-15)

- 1 Full Syllabus
- 2 Marks: 40

Grading System

Marks Range	Grade
90-100	A+
80-89	A
70-79	B
60-69	C
50-59	D
<50	F